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AHEAD Online

Computer Networks

Objective

- Introduction to Computer Networks.
- TCP/IP Protocol Suit.
- Network Architectures:
  - Peer-to-Peer Architecture, and
  - Client-Server Architecture.

Computer Networks

- The term 'Network' means a group, chain or a collection of something that come together for the purpose of communication.
- A computer network is a group of related computers that exchange resources, files and possess some kind of communications within themselves.

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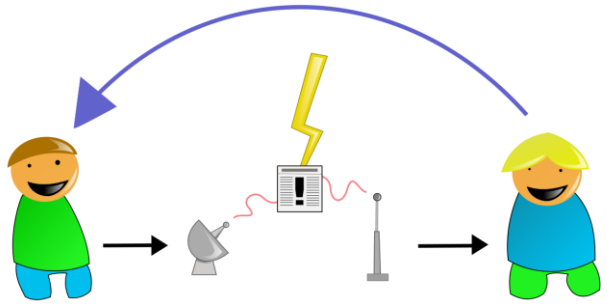
Advantages of Computer Networks

- File Sharing

- Flexible Access

- Better Communication



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- Resource Sharing

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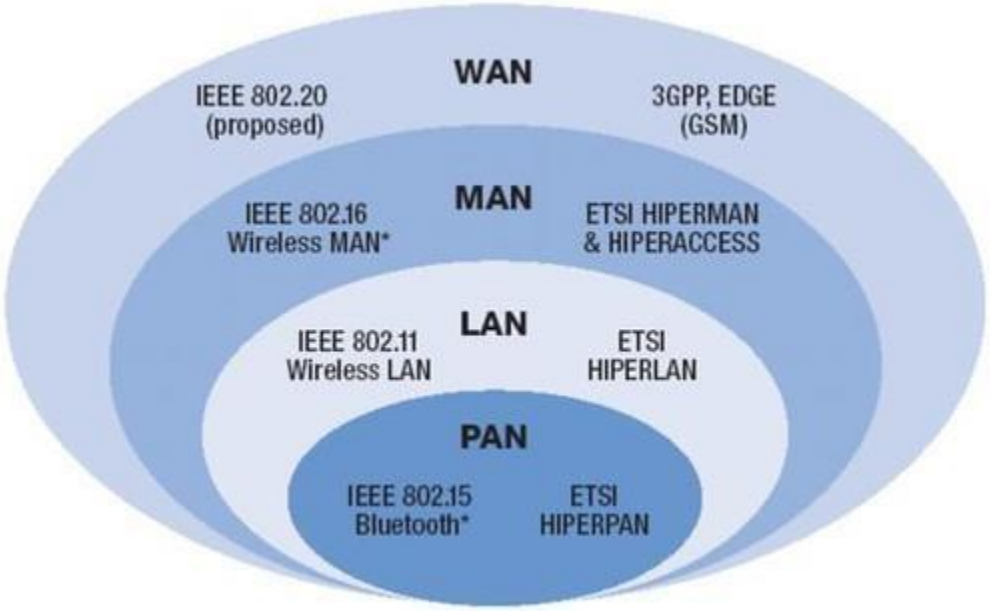
Types of Computer Network

- Computer Networks are classified based on:
  - Size - Geographic Area
  - Ownership - Public domain or owned an individual or a corporation
  - Topology - Arrangement of Devices

Classification Based On Size

- Computer Networks based on Size are:
  - Personal Area Network (PAN)
  - Local Area Network (LAN)
  - Metropolitan Area Network (MAN)
  - Wide Area Network (WAN)



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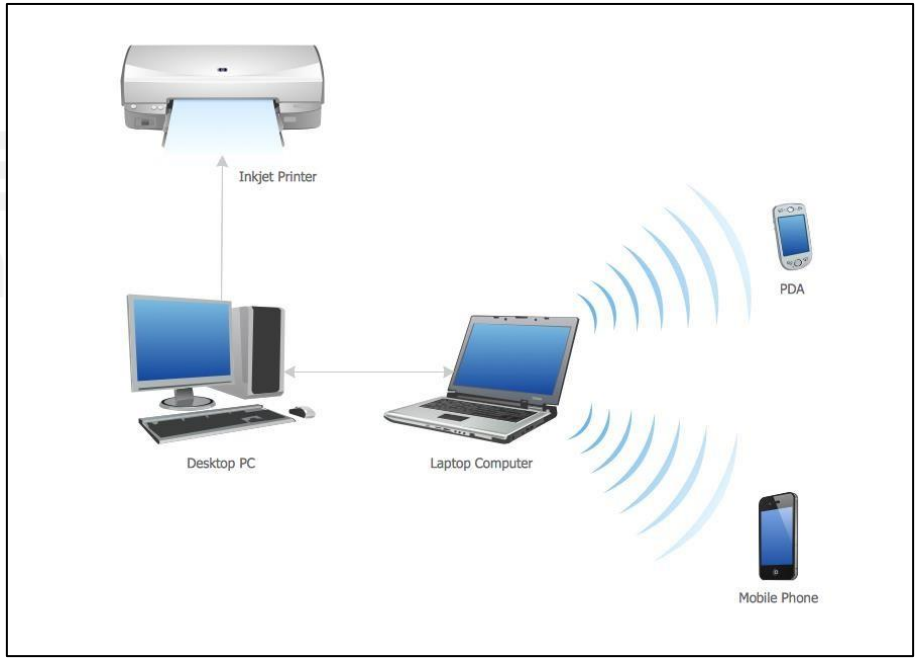
Personal Area Network (PAN)

- A personal area network (PAN) is a computer network organized around an individual for personal use only.
- They typically involve a computer, phone, printer, tablet, or some other device like a PDA.
- A PAN may be wireless or carried over wired interfaces such as USB.



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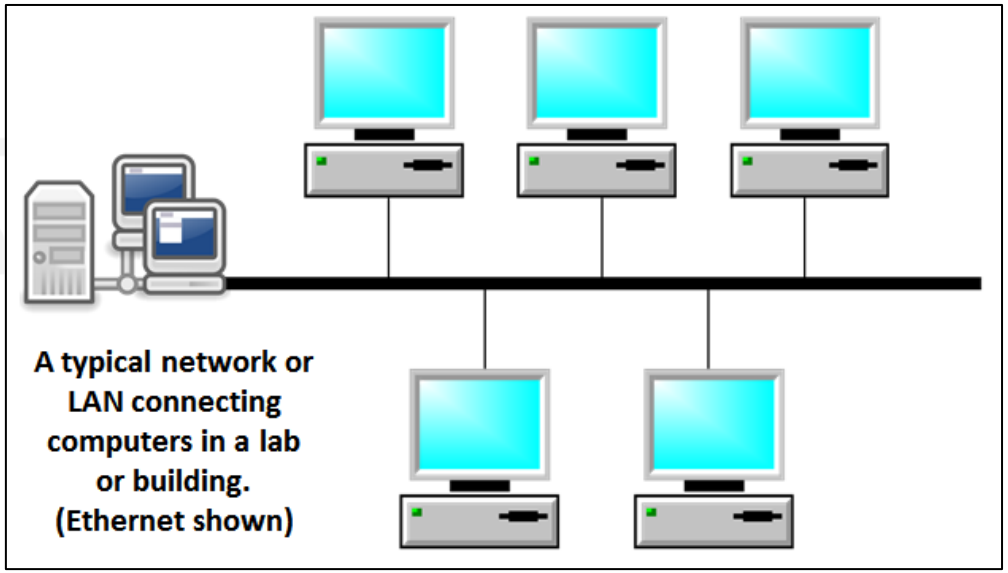
Local Area Network (LAN)

- A local area network (LAN) is a computer network that interconnects computers within a limited area such as a residence, school, laboratory, university campus or office building.



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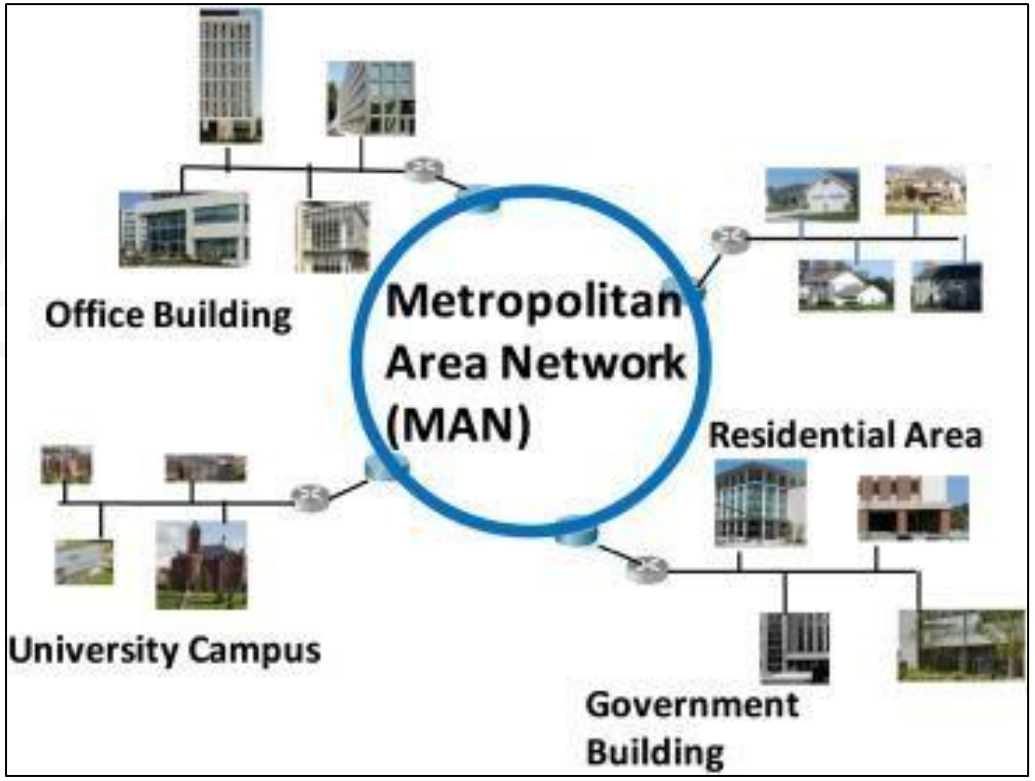
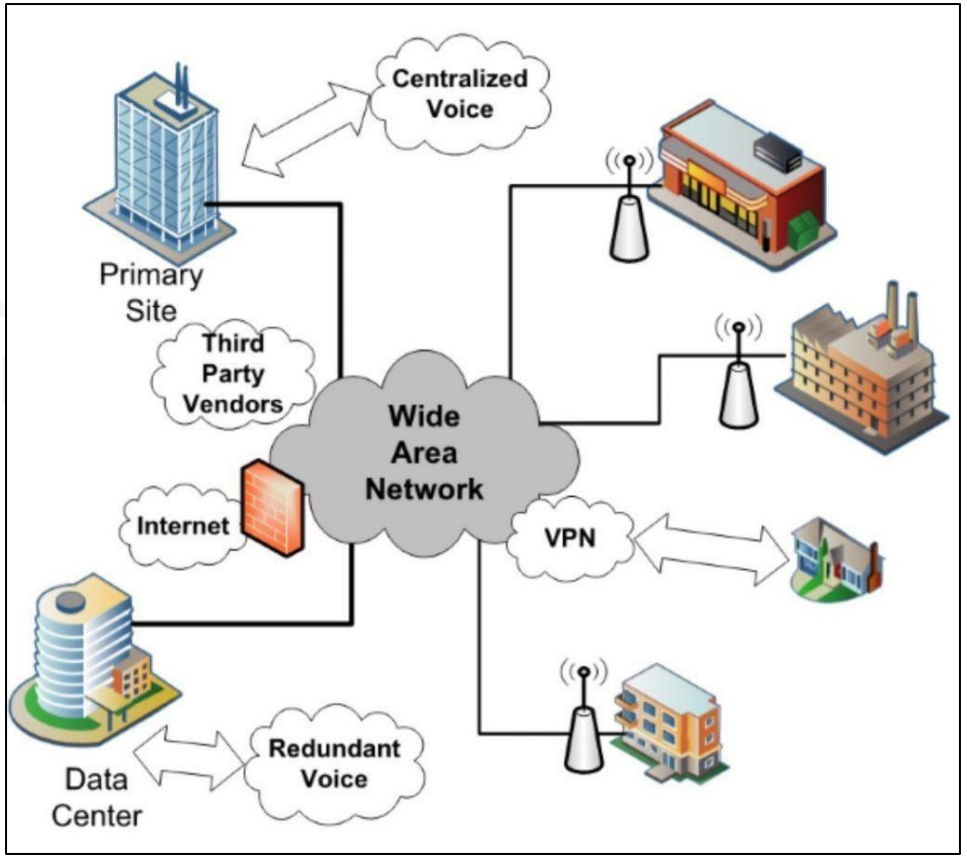
Metropolitan Area Network (MAN)

- A MAN is a network that interconnects computers within a metropolitan area such as a city.
- MAN is larger than LAN but smaller than WAN.



Wide Area Network (WAN)

- A wide area network (WAN) is a computer network that covers a large geographical area.
- It may comprise a region, a country, a continent or even the whole world.



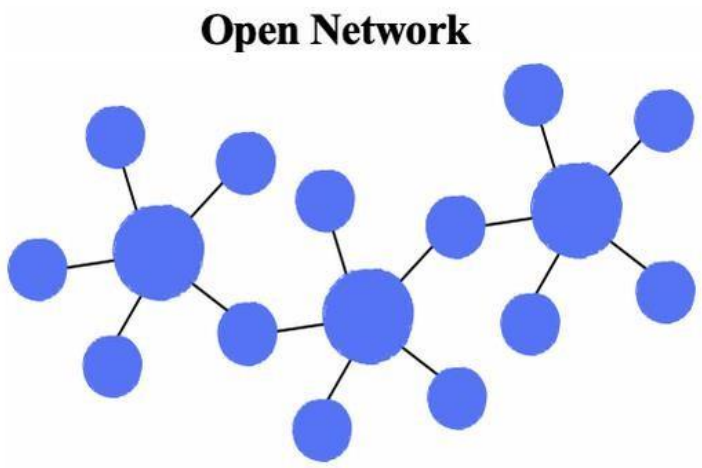
Classification Based On Ownership

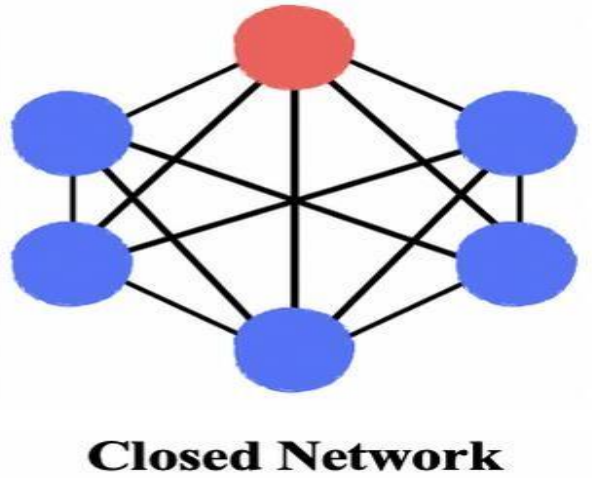
Open Network

- If the network's internal operation is in the public domain.

Closed Network

- Private network that can only be used by authenticated, authorized devices.



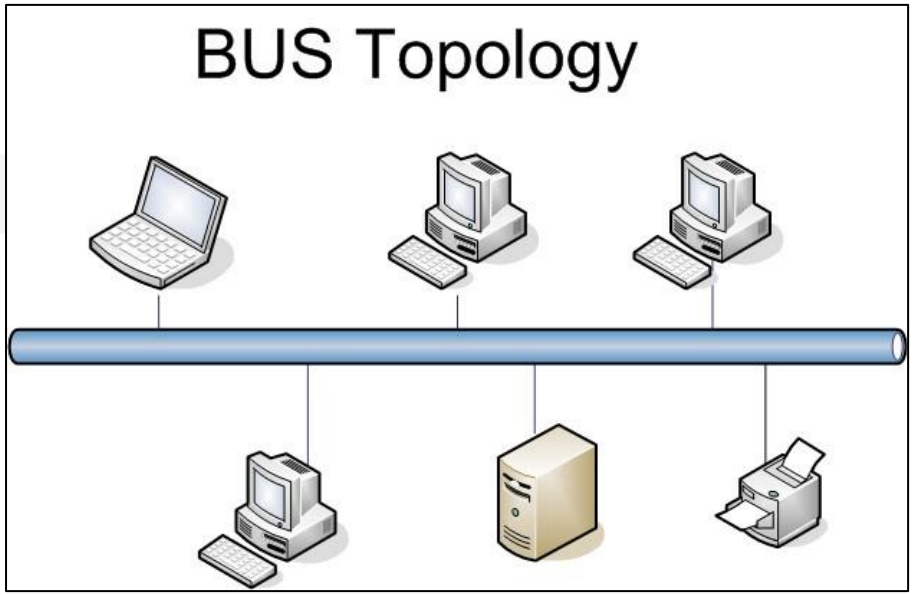


Classification Based On Topologies

- Topology defines the structure of the network of how all the components are interconnected to each other.
- Types are
  - Bus
  - Ring
  - Star
  - Mesh
  - Hybrid

Bus Topology

- It is the simplest network topology
- All computers and networking devices are connected to a single Backbone Cable.



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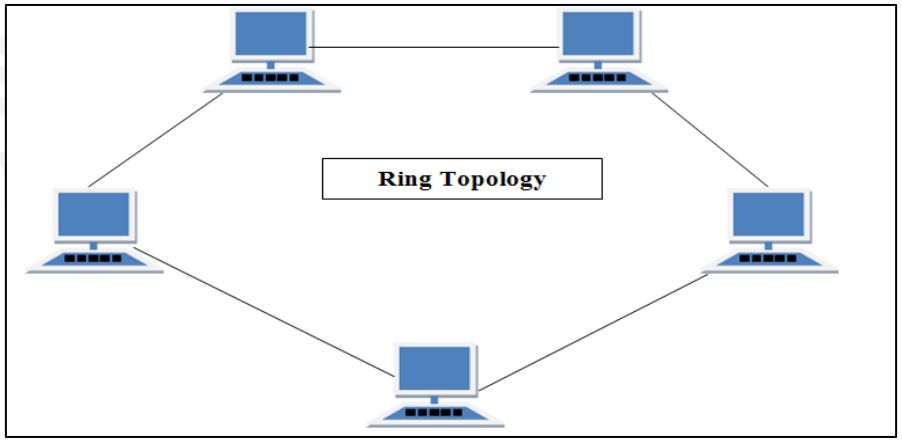
Ring Topology

- Every computer is connected to another computer on each side.
- This allows for each computer to have exactly two neighboring computers.
- Data is transmitted in the clockwise direction from one node to another.



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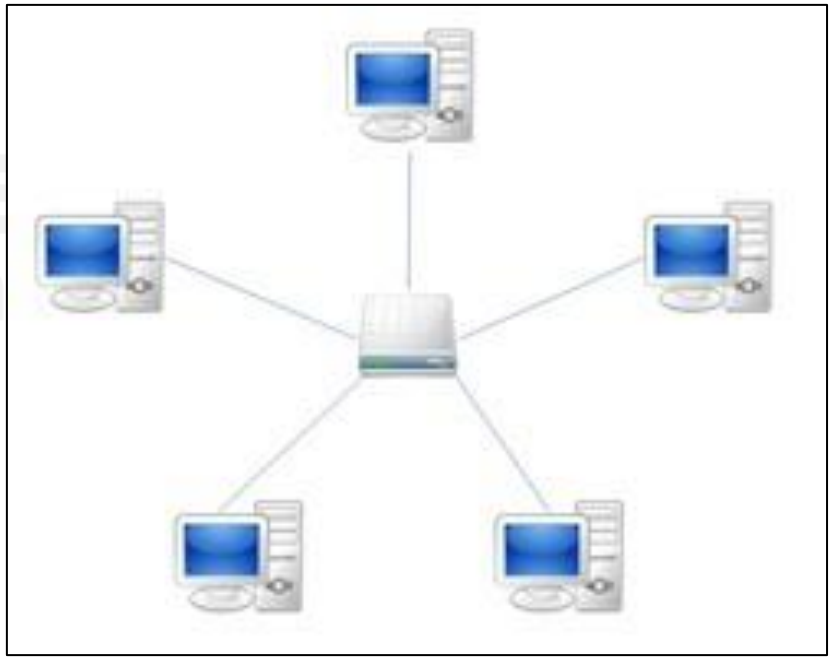
Star Topology

- In the star topology, all the nodes are connected to a common device known as hub.
- Hub manages the overall data transmission in the network.



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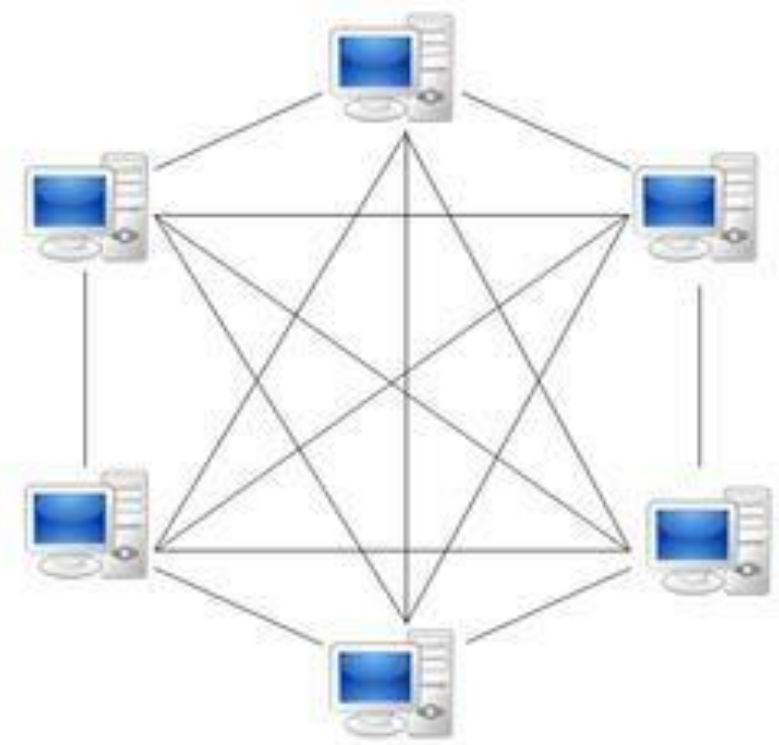
Mesh Topology

- In mesh topology, each computer is connected to every other computer in point-to-point mode.
- If we have n computers, we must have n(n-1)/2 links.



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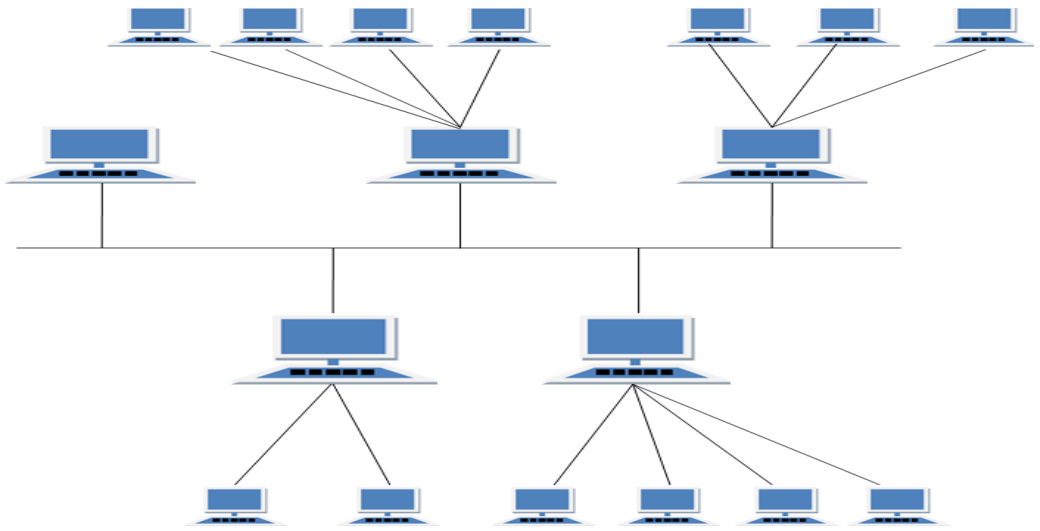
Hybrid Topology

- The hybrid topology is the combination of multiple topologies.
- The hybrid topology is created when two different network topologies are interconnected.



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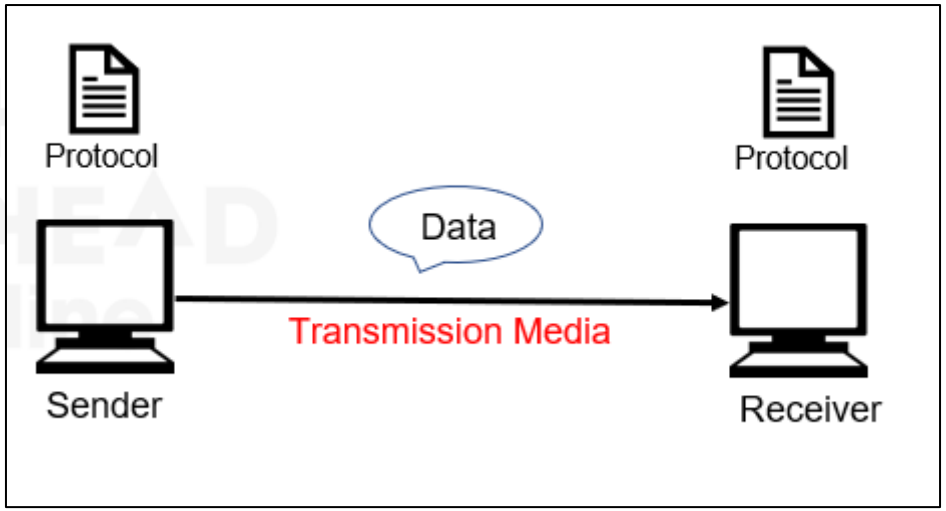
Transmission Media

- Transmission media refer to the media through which data can be carried from a source to a destination.
- Data is transmitted from one device to another through electromagnetic signals.



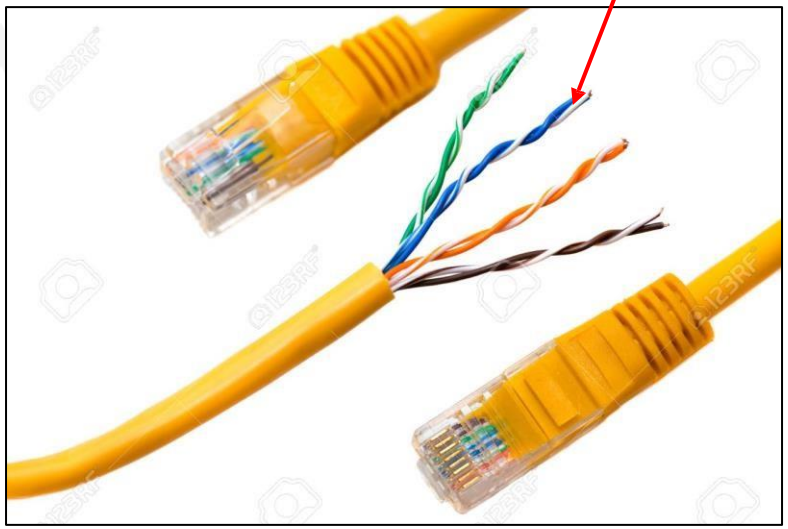
Types of Transmission Media

- Guided/ Wired Medium: Uses cables or wires to transmit data.
  - Twisted Pair Cable
  - Coaxial Cable
  - Fiber Optic Cable
- Unguided/ Wireless Medium: Transmit data through electromagnetic waves without using any physical medium (wireless transmission).
  - Electromagnetic Waves
  - Radio Waves
  - Micro Waves
  - Infrared Waves

Twisted Pair Cable

- Twisted pair cable is the most popular network cable for data transmission.
- It is lightweight, easy to install, inexpensive and supports data speed up to 100Mbps.
- It is a pair of copper wires (each pair of cable twisted together to form a single media) is most commonly used wire for transmitting signals due to good performance at low cost.
- A twist between wires help in reducing noise and crosstalk.

Twisted Pair



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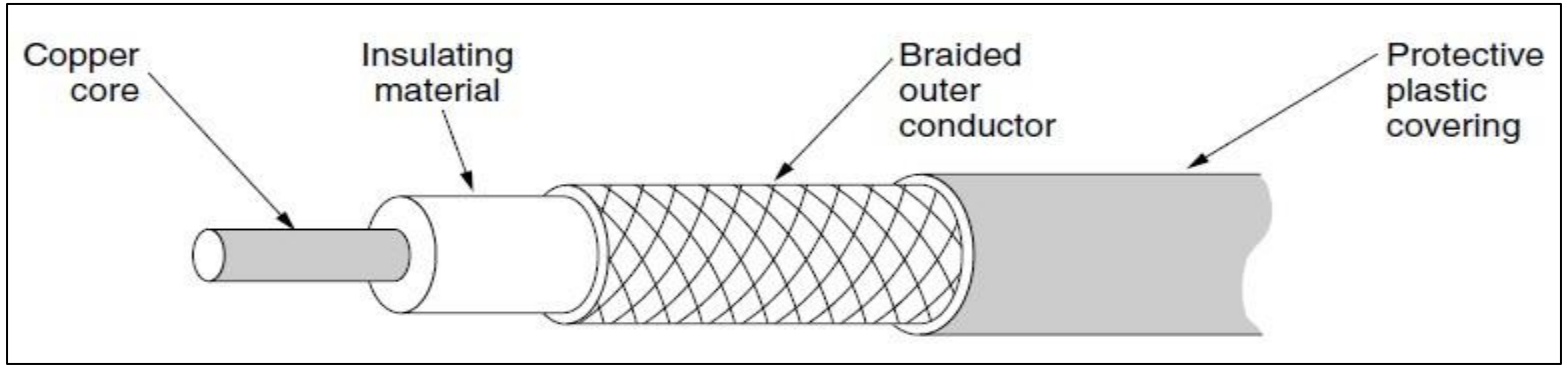
Coaxial Cable

- Coaxial cable has two wires of copper.
- The core/inner copper wire in the center is made of solid conductor which is actually used for data transmission. It is enclosed in an insulating sheath.
- The second/ external copper wire is wrapped around which is used to protect against external electromagnetic interference (noise).



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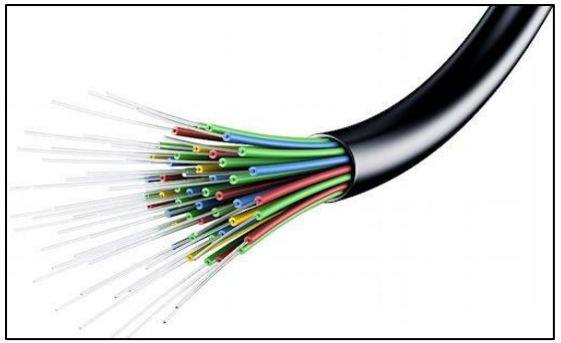
Fiber Optic Cable

- A fiber optic cable is made of high quality of thin glass or plastic which is used to transfer digital data signals in the form of light over thousands of miles.
- It is not affected by electromagnetic interference.
- It provides high data transmission and designed for long-distance.
- It transmit data signals using light pulses generated by small lasers or LEDs.



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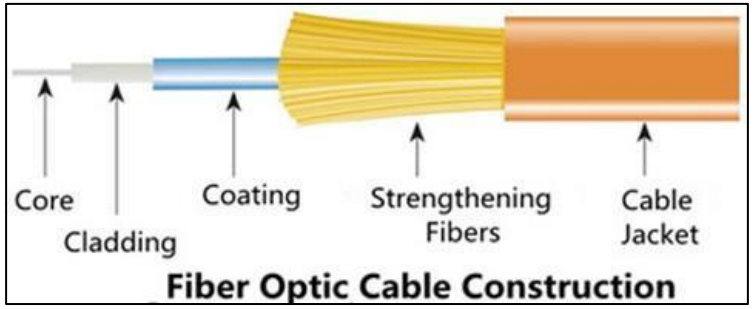

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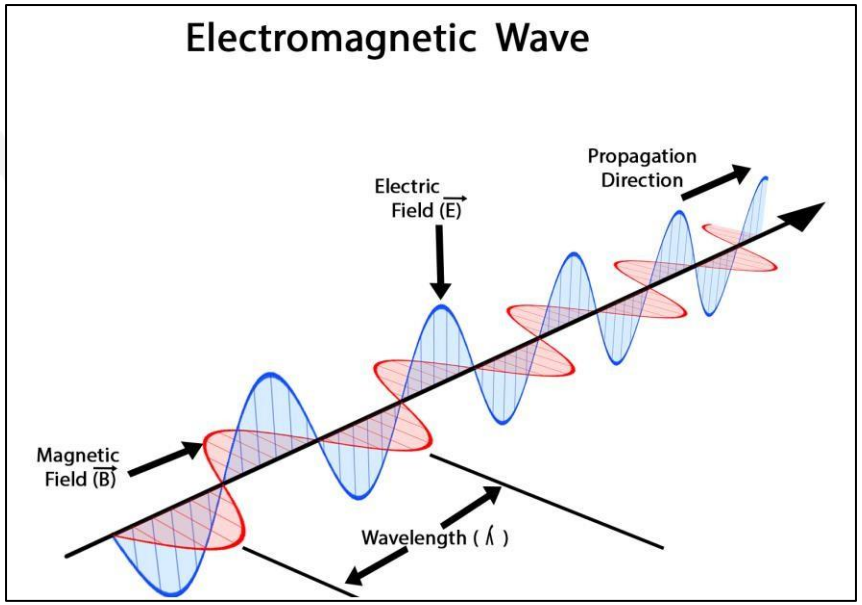
Electromagnetic Wave

- Electromagnetic waves are formed when an electric field comes in contact with an magnetic field.
- EM waves travels with a constant velocity of 3.00 x 108 ms^{-1} in vacuum.
- They can travel through anything like air, solid material, or vacuum.



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Radio Waves

- Radio waves are electromagnetic waves which are the longest wavelengths in the electromagnetic spectrum.
- Its wavelength can be between 1mm and 10,000km and in frequencies can be as high as 300 GHz to as low as 30Hz.
- Transmitters are used to generate radio waves artificially which are received by radio receivers using antennas.
- Radio waves are omnidirectional.

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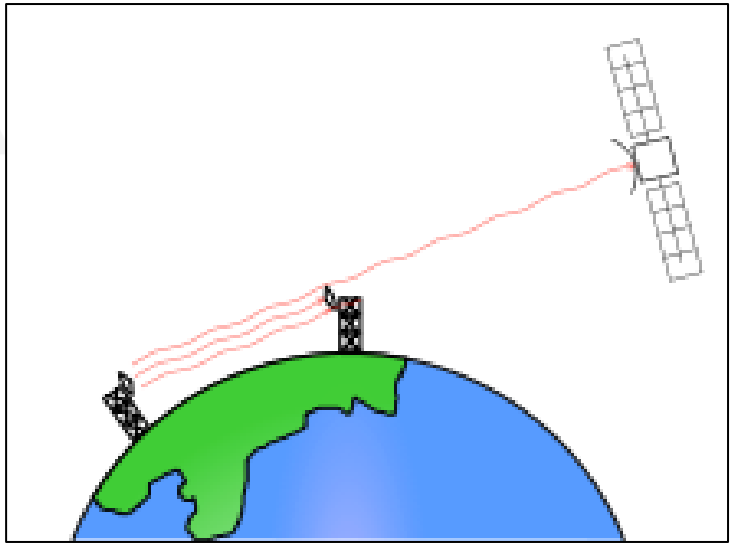
Micro Waves

- Microwave is a type of radio wave with high frequencies.
- The frequency range of microwave is between from 300MHz to 300GHz.
- Microwaves are unidirectional.
- Used for satellite communication, navigation, radar, remote sensing and other short range communication systems.



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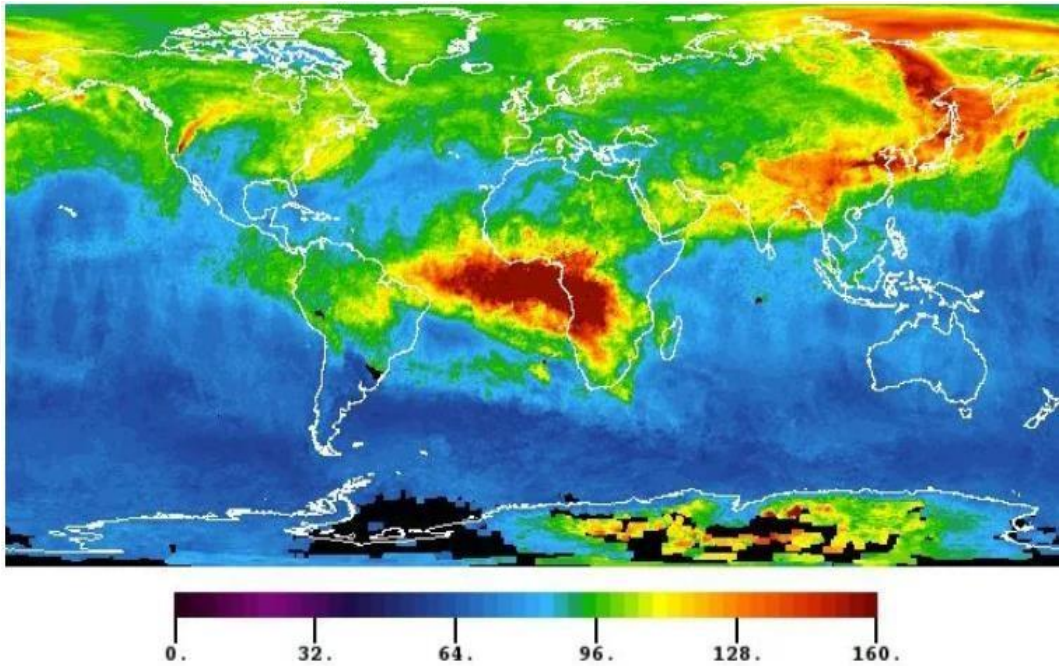
Infrared Waves

- Infrared waves are used for very short range communication like TV remote, wireless mouse, wireless printer etc.
- It has very high frequency, these waves cannot penetrate walls.
- It also has very high data transfer rate.



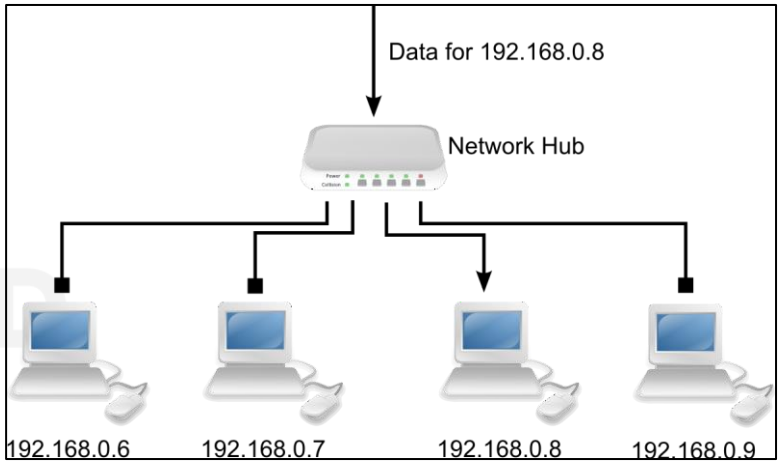
An image of Earth in infrared wavelengths shows relative temperatures around the world. The photo includes a plume of carbon monoxide pollution near the Rim Fire that burned near Yosemite National Park in California on Aug. 26, 2013. (Image credit: NASA/JPL-Caltech/Space Science Institute)

Network Devices

- Network devices, or networking hardware, are physical devices that are required for communication and interaction between hardware on a computer network.
- Types of network devices:
  - Hub
  - Switch
  - Router
  - Bridge
  - Repeater

Hub

- Hub is a network hardware device for connecting multiple devices together.
- When computer requests for some information it first sends the request to the Hub through cable in a network.
- Hub will broadcast this request to the entire network.
- All the devices will check whether the request belongs to them or not. If not, the request will be dropped.



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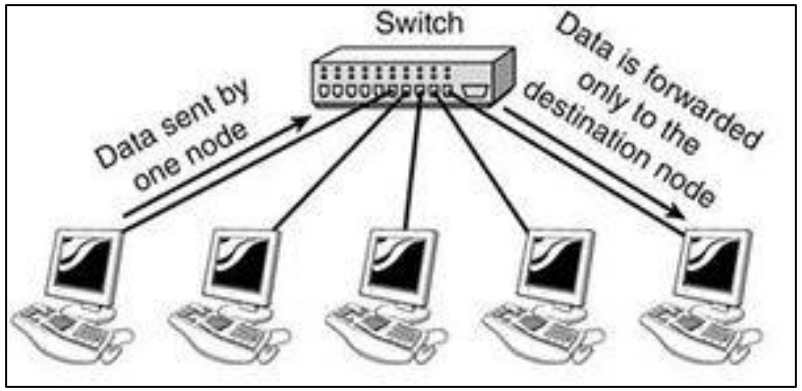
Switch

- Switches generally have a more intelligent role than hubs.
- A switch is a multiport device that improves network efficiency.
- The switch maintains limited routing information about nodes in the internal network, and it allows connections to systems like hubs or routers.
- Generally, switches can read the hardware addresses of incoming packets to transmit them to the appropriate destination.



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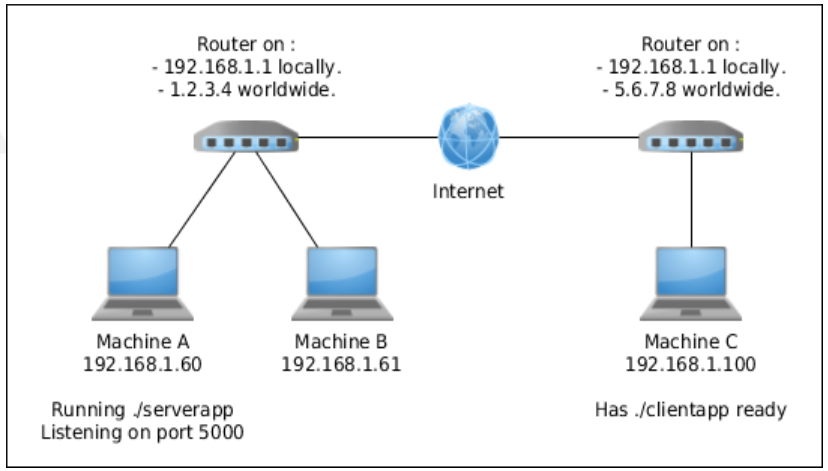
Router

- Routers help transmit packets to their destinations by charting a path through the sea of interconnected networking devices using different network topologies.
- They establish communication by maintaining tables about destinations and local connections.
- A router contains information about the systems connected to it and where to send requests if the destination isn't known.



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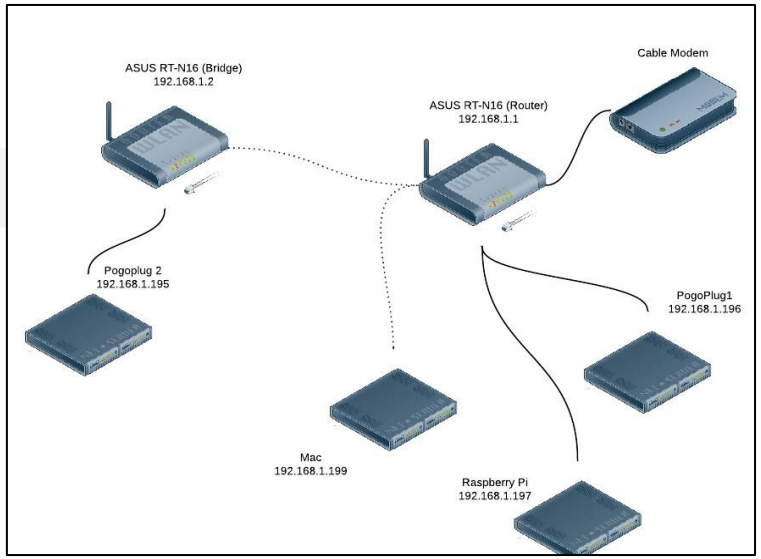
Bridge

- Bridges are used to connect two or more hosts or network segments together.
- The basic role of bridges in network architecture is storing and forwarding frames between the different segments that the bridge connects.
- They use hardware Media Access Control (MAC) addresses for transferring frames.
- By looking at the MAC address of the devices connected to each segment, bridges can forward the data or block it from crossing.
- Bridges can also be used to connect two physical LANs into a larger logical LAN.



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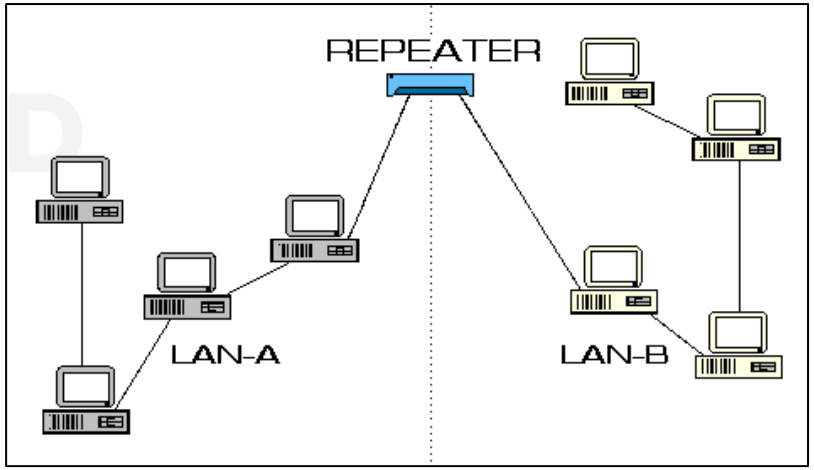
Repeater

- A repeater is an electronic device that amplifies the signal it receives.
- You can think of repeater as a device which receives a signal and retransmits it at a higher level or higher power so that the signal can cover longer distances, more than 100 meters for standard LAN cables. Repeaters work on the Physical layer.



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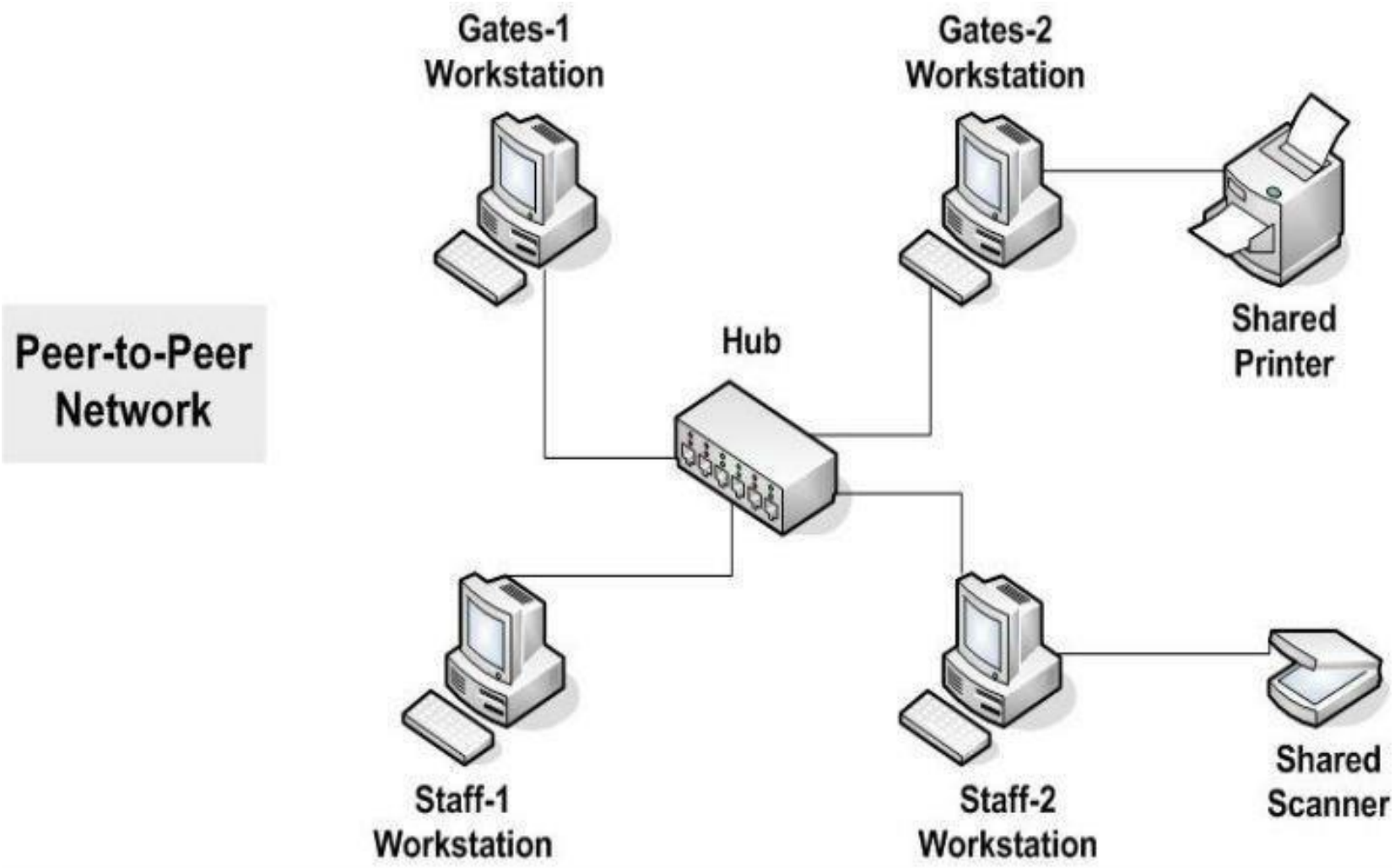
Network Architecture

- Network architecture is the design of a computer network.
- It is the logical and structural layout of the network, consisting of transmission equipment, software and communication protocols, and infrastructure (i.e. wired or wireless) transmission of data and connectivity between components.
- The two types of widely used network architectures are:
  - peer-to-peer or P2P, and
  - client/server or tiered.

Peer-to-Peer Architecture

- Peer-to-peer networks involve two or more computers pooling individual resources.
- The shared resources are available to every computer in the network.
- Each computer acts as both the client and the server, communicating directly with the other computers.
- For example, a printer on one computer can be used by any other computer on the network.
- P2P networks are inexpensive to set up. All you need is a way to connect them, like an Ethernet cable or a Wi-Fi router.

Peer-to-Peer Model



Advantages of Peer-to-Peer Network

- Does not require a dedicated server which means its less costly.
- If one computer stops working, the other computers connected to the network will continue working.
- Installation and setup is quite painless because of the built-in support in modern operating systems.

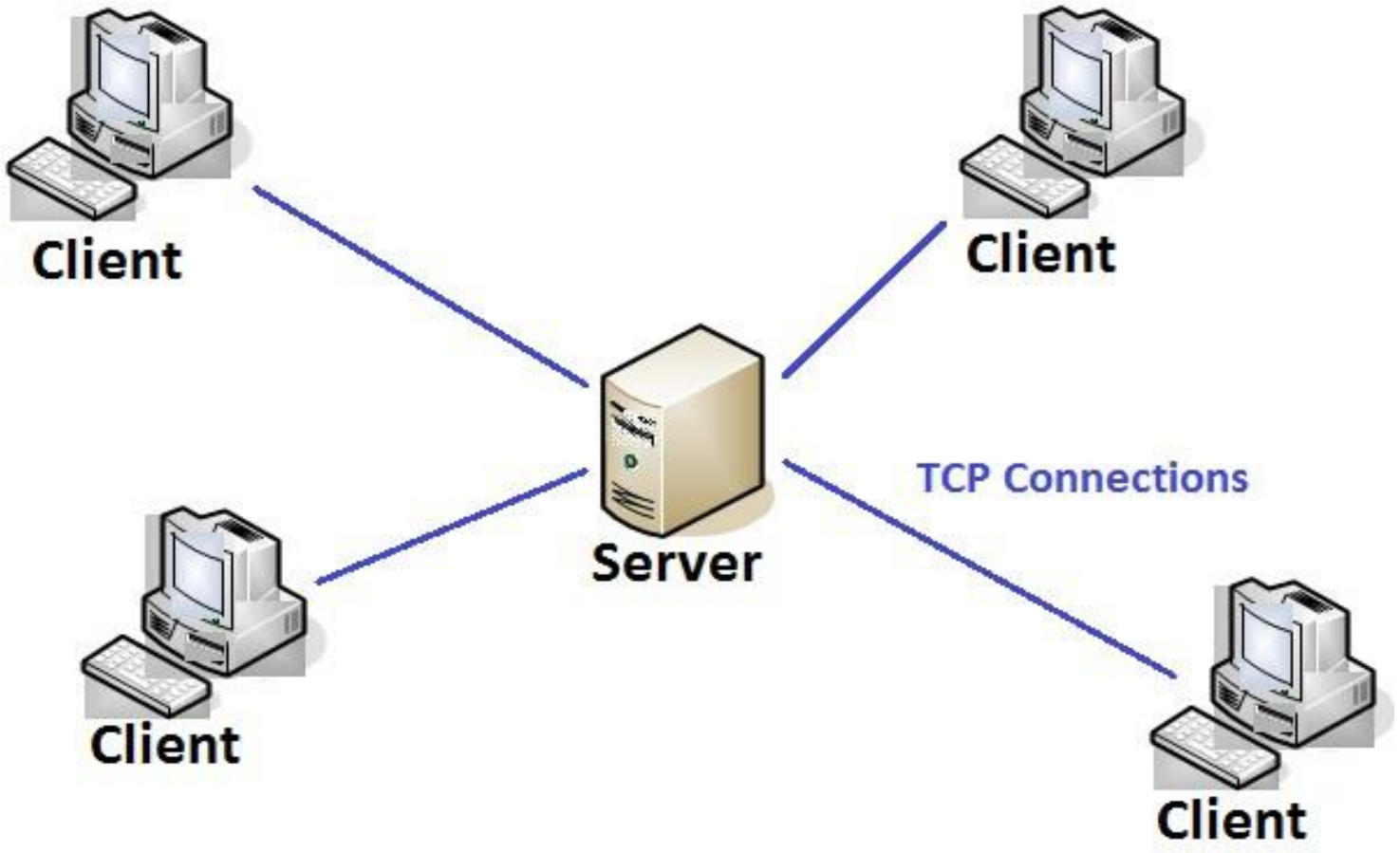
Disadvantages of Peer-to-Peer Network

- Security and data backups are to be done to each individual computer.
- As the numbers of computers increases on a P2P network, the performance, security, and access becomes more complex.

Client-Server Architecture

- Client-server architecture, architecture of a computer network in which many clients (remote processors) request and receive service from a centralized server (host computer).
- In a client/server network, a centralized, really powerful computer(server) acts as a hub in which other computers or workstations(clients) can connect to.
- This server is the heart of the system, which manages and provides resources to any client that requests them.

Client-Server Model



Advantages of Client-Server Network

- Resources and data security are controlled through the server.
- Not restricted to a small number of computers.
- Server can be accessed anywhere and across multiple platforms.

Disadvantages of Client-Server Network

- Can become very costly due to the need of a server as well as networking devices such as hubs, routers, and switches.
- If and when the server goes down, the entire network will be affected.
- Technical staff needed to maintain and ensure network functions efficiently.

Layered Tasks

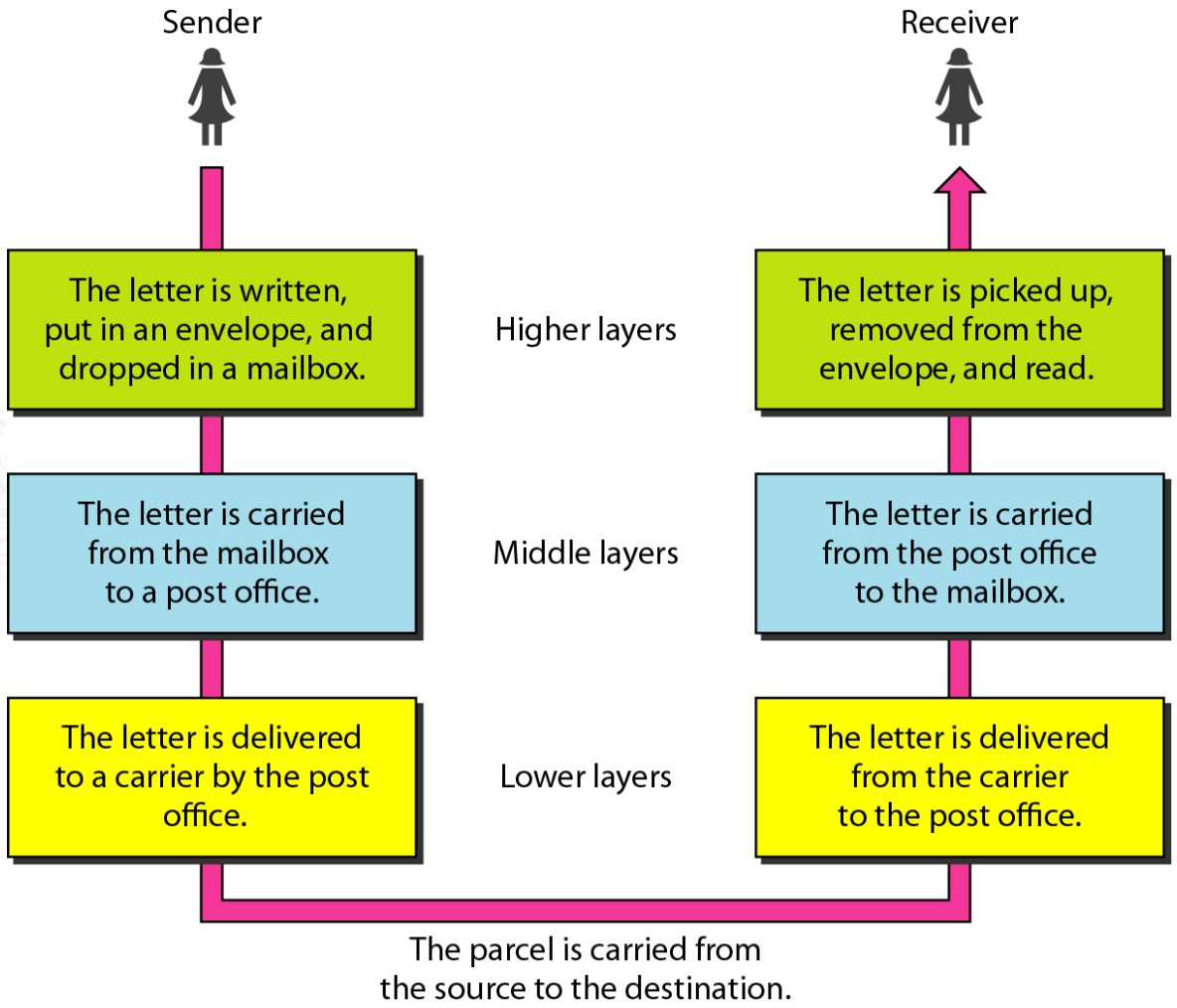
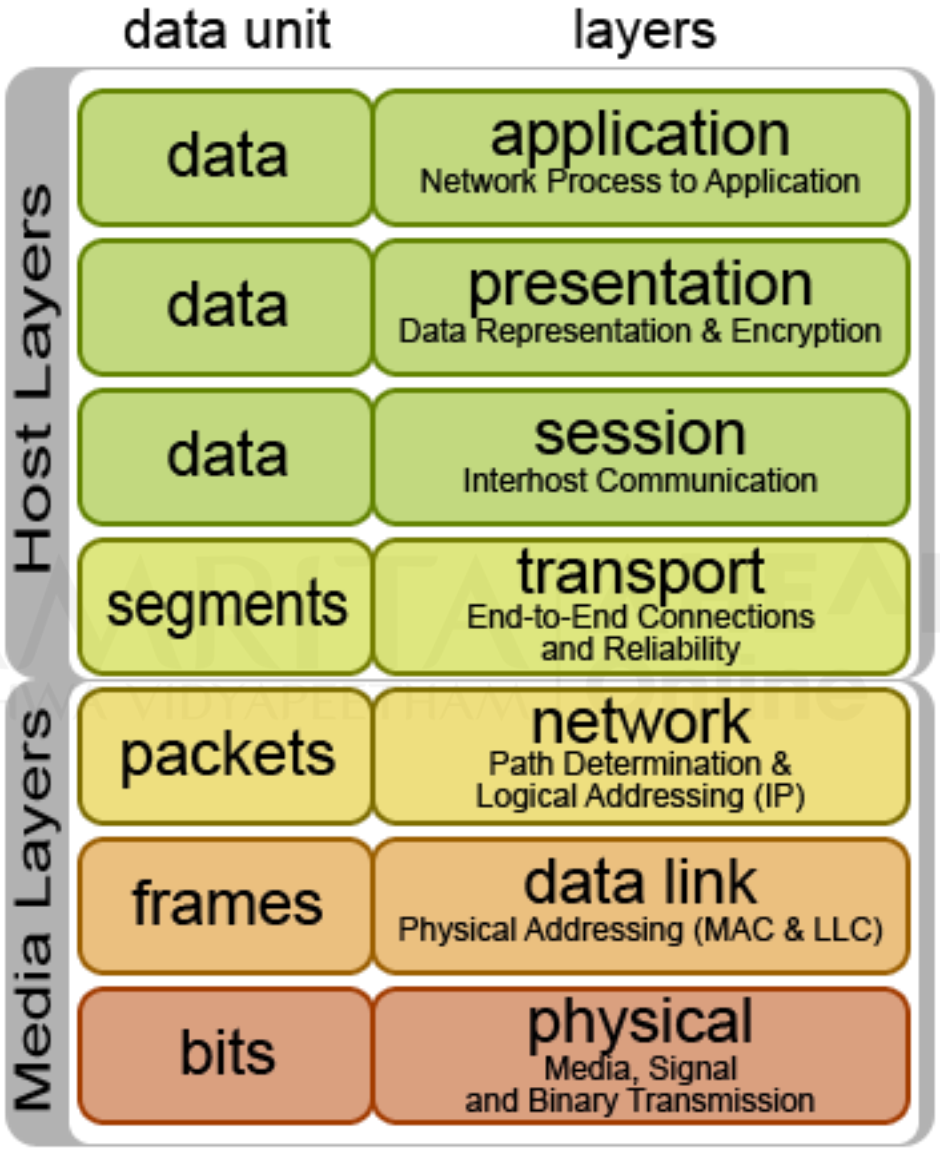


Image Courtesy: Data Communication and Networking Behrouz A. Forouzan, McGraw-Hill

Layered Tasks

- Hierarchy of tasks.
- Sender.
- Receiver.
- Carrier.

OSI Model



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Organization of Layers

- Network Support Layers
  - Physical, data link and network layers.
  - Deals with physical aspects of moving data from one device to another.
- User Support Layers
  - Session, Presentation and application layer.
  - Provides interoperability among different software systems.
- Transport layer links these two subgroups.

Physical Layer

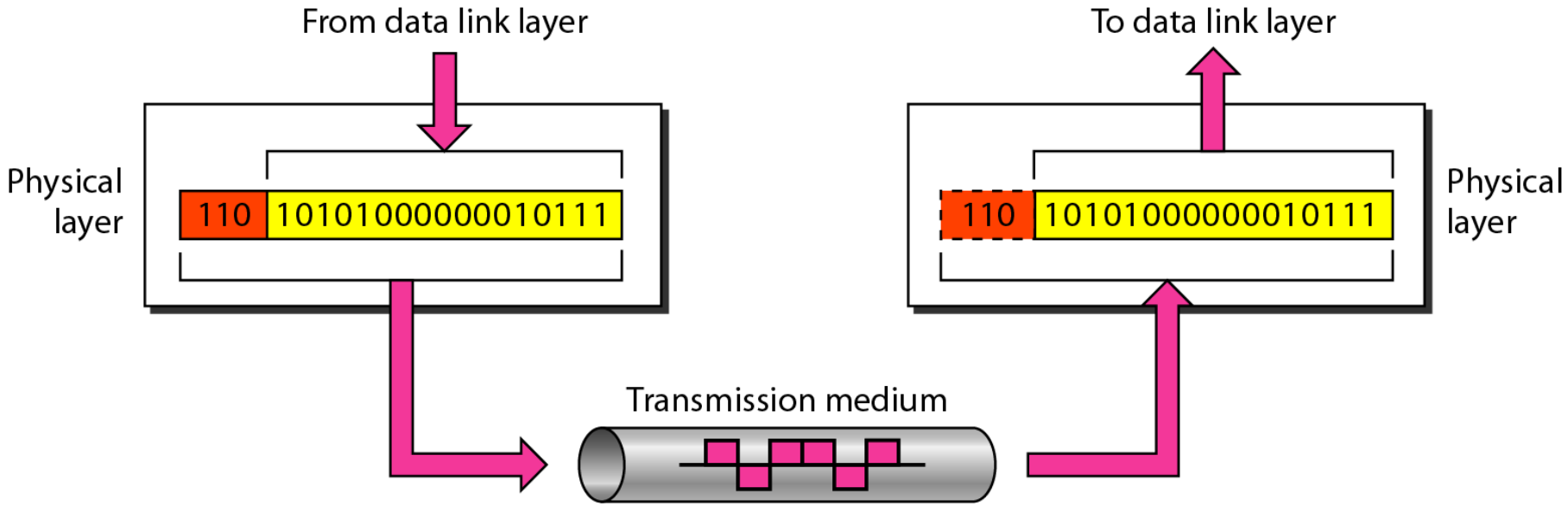


Image Courtesy: Data Communication and Networking Behrouz A. Forouzan, McGraw-Hill

Data Link Layer

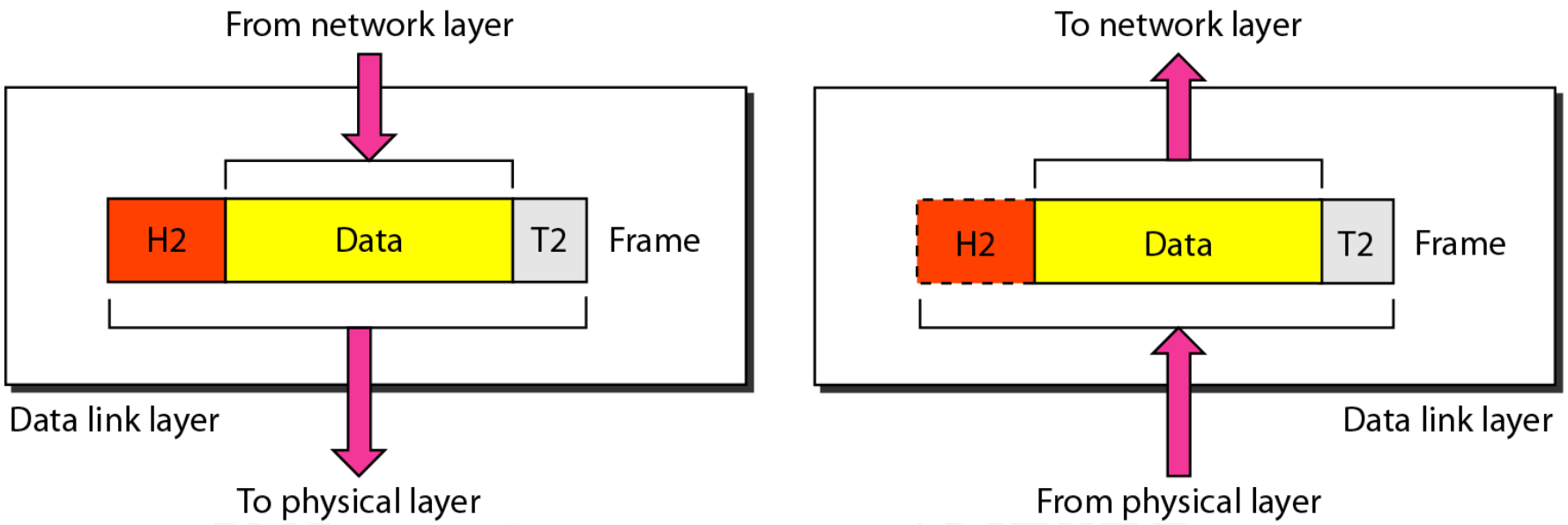
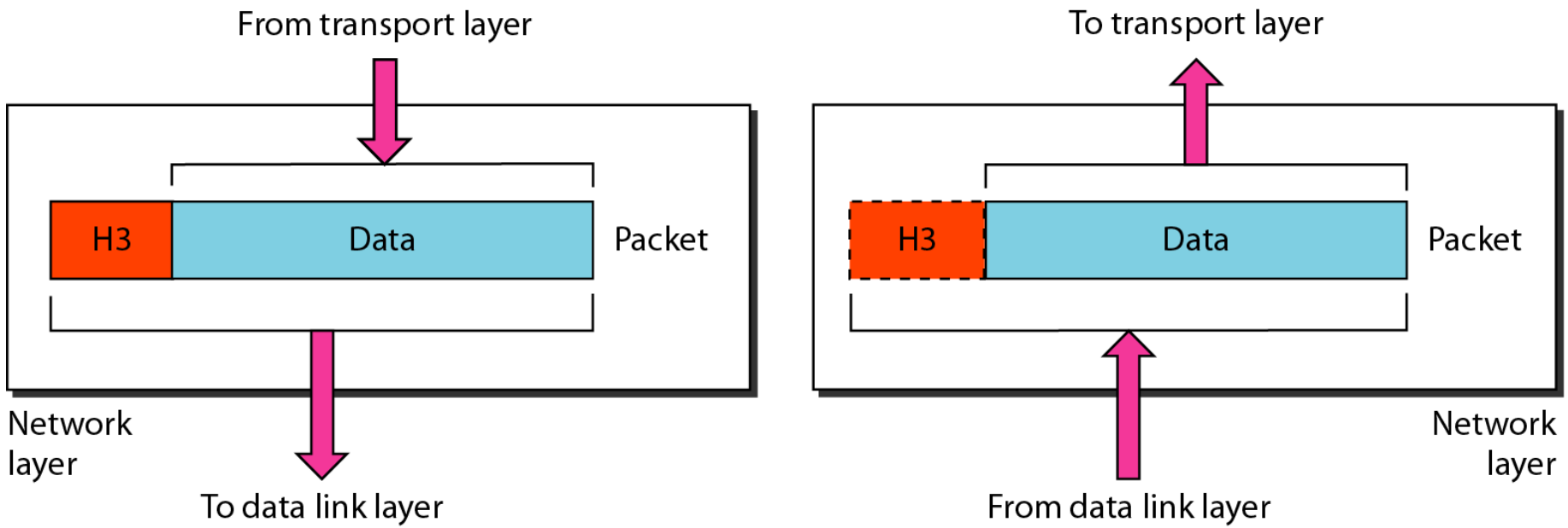


Image Courtesy: Data Communication and Networking Behrouz A. Forouzan, McGraw-Hill

- Responsible for moving frames from one node to another.
- Also performs error control and flow control.

Network Layer



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Transport Layer

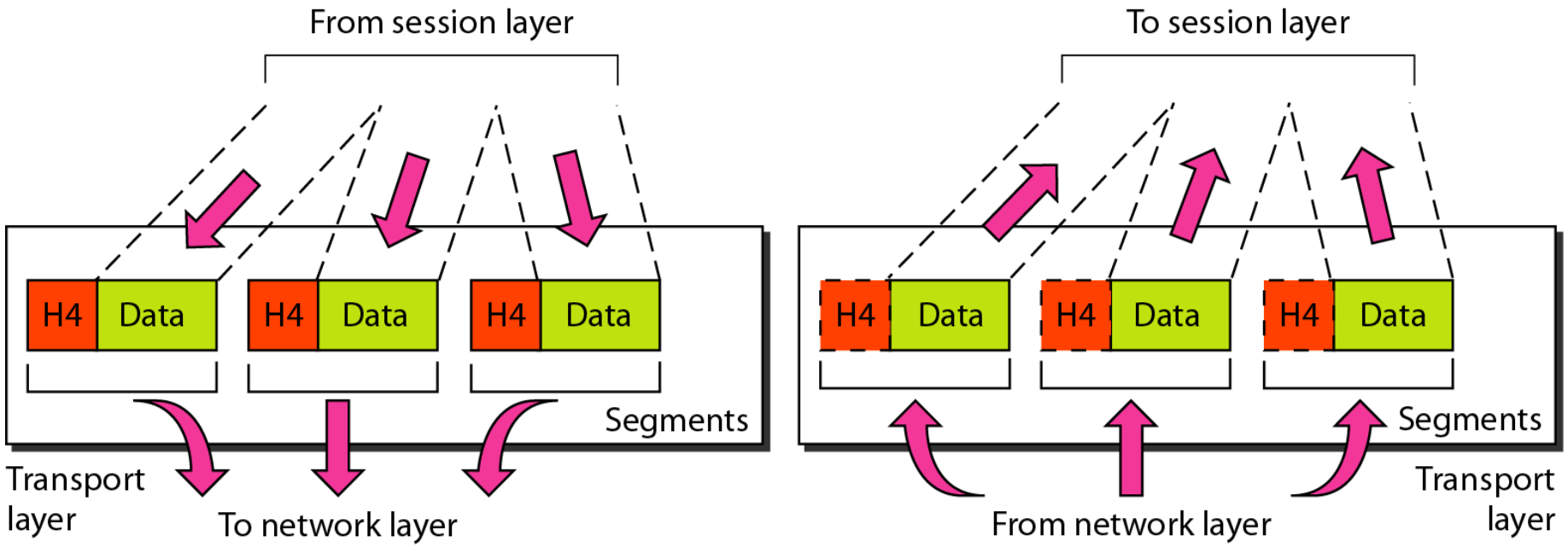
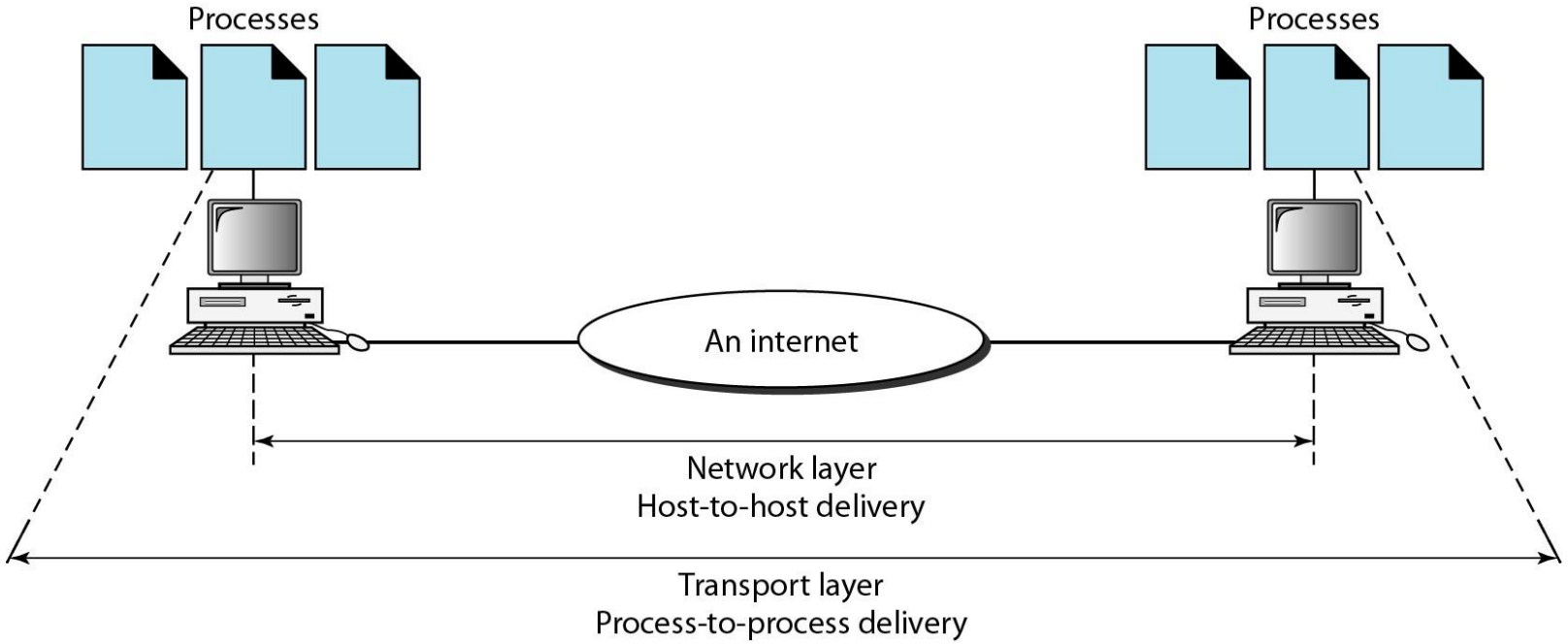


Image Courtesy: Data Communication and Networking Behrouz A. Forouzan, McGraw-Hill

Process to Process Delivery



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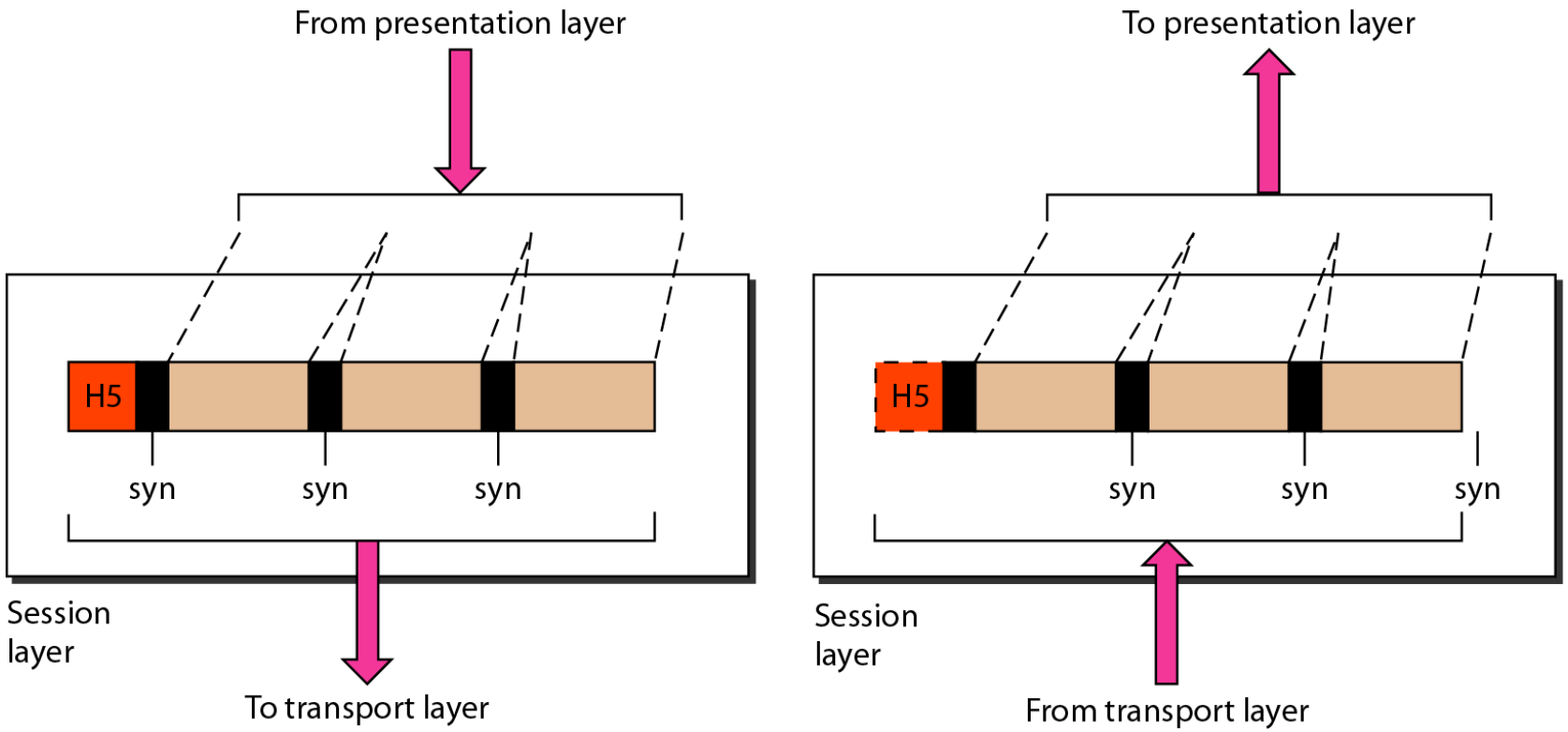
Session Layer



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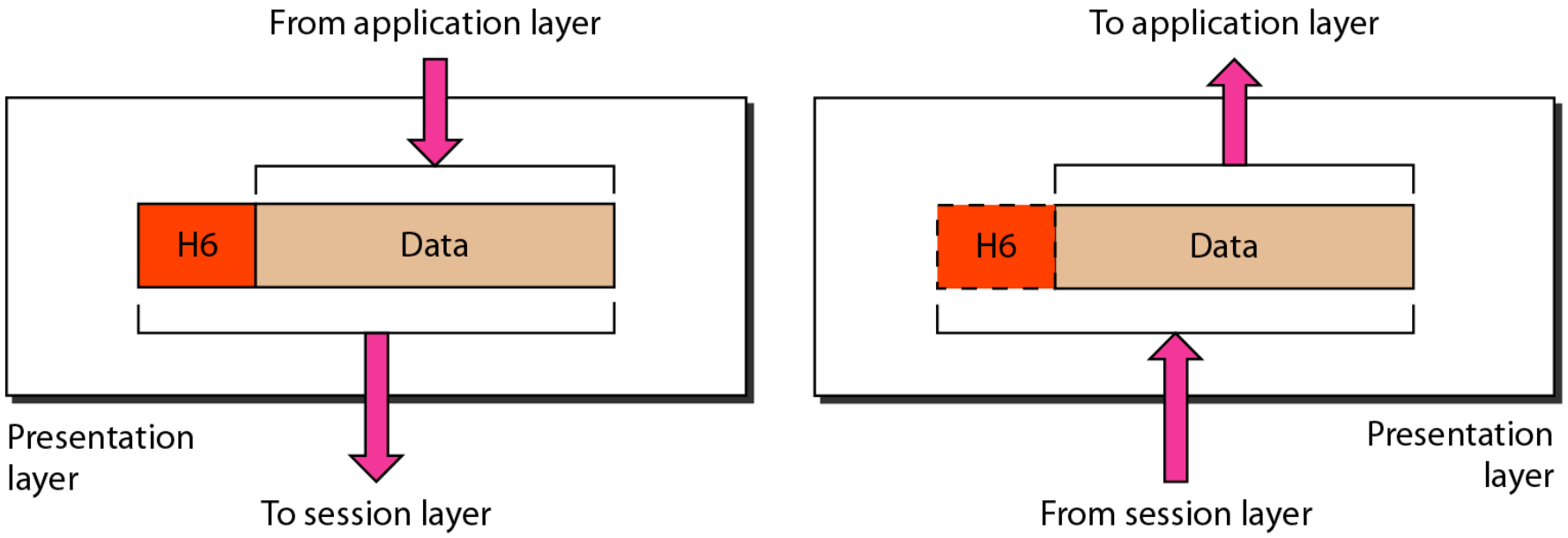
- It establishes, maintains, and synchronizes the interaction among communicating systems.

Presentation Layer



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Application Layer

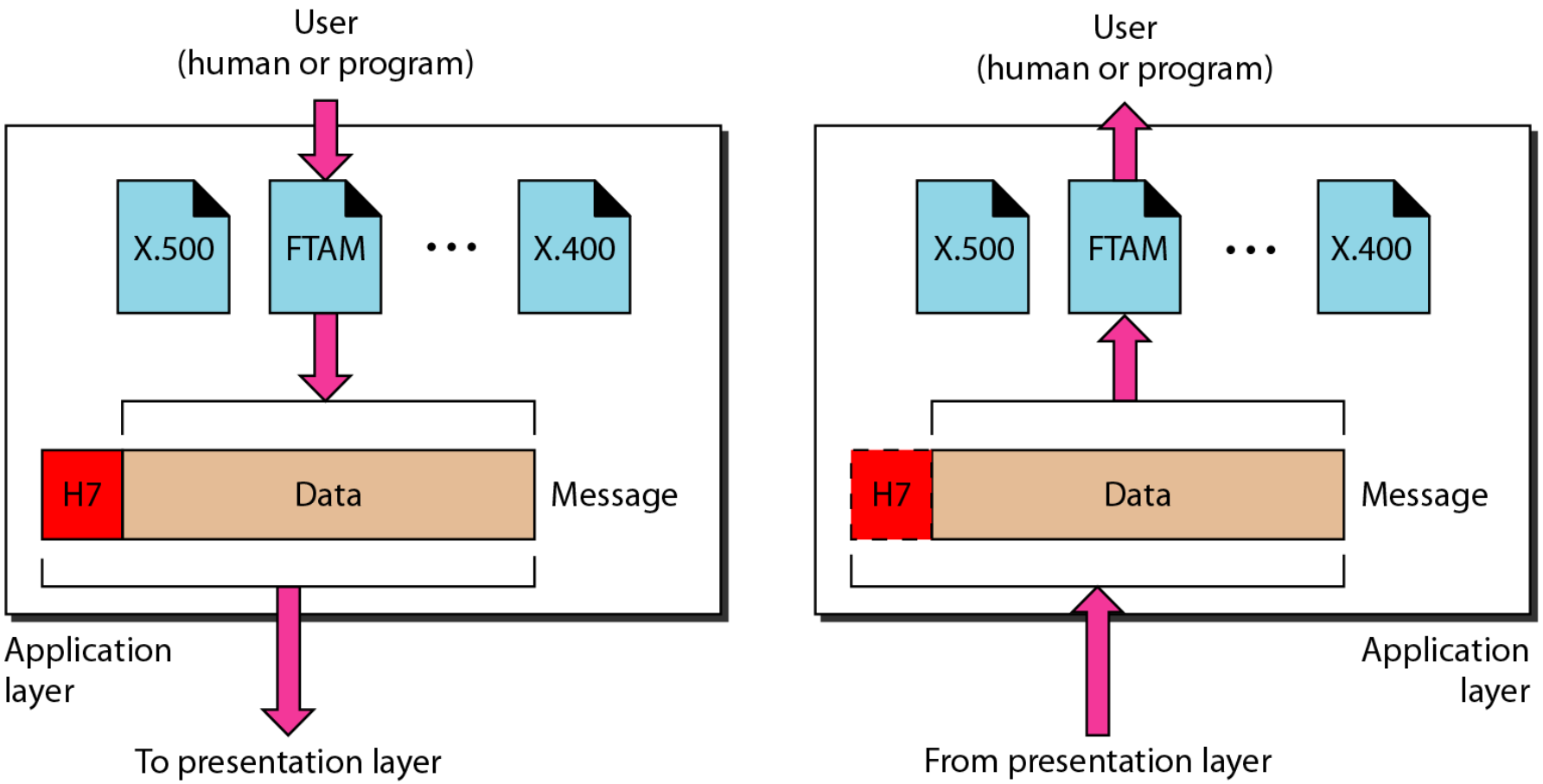
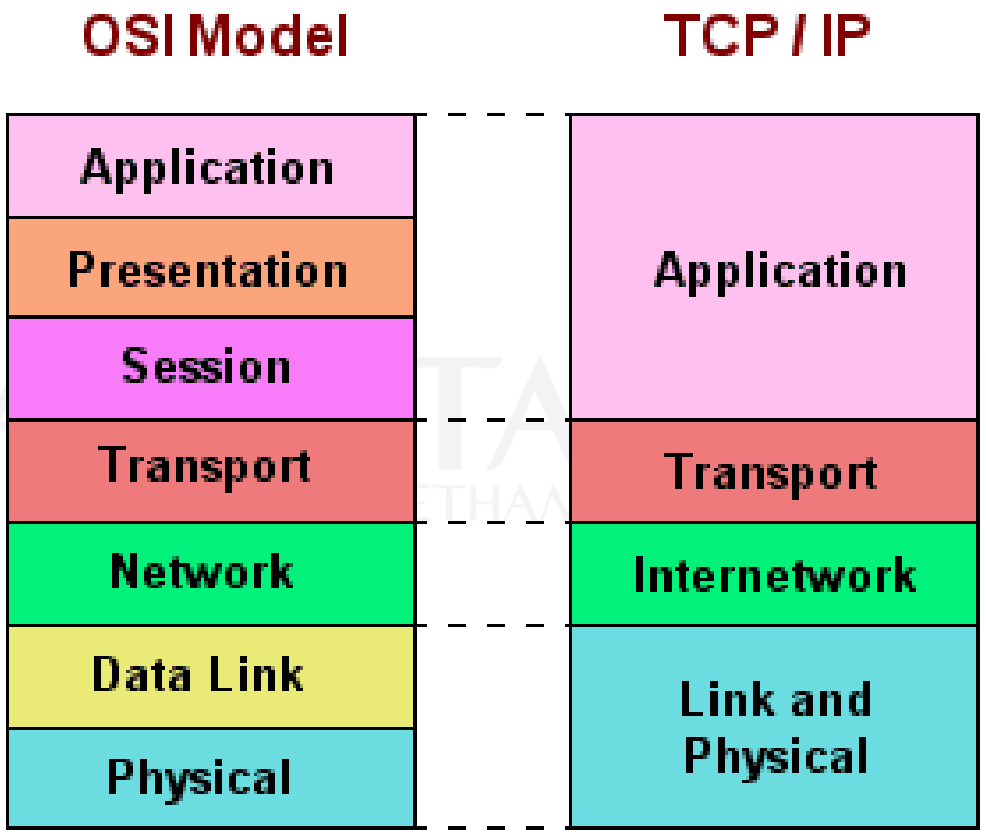


Image Courtesy: Data Communication and Networking Behrouz A. Forouzan, McGraw-Hill

Organization of Layers

- Host to network Layer(Link and Physical).
- Internet Layer.
- Transport layer.
- Application Layer.

TCP and OSI Model



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TCP/IP

- TCP/IP is a hierarchical protocol.
- Each upper layer protocol is supported by one or more lower layer protocols.
- Transport Layer Protocols
  - Transmission Control Protocol(TCP).
  - User Datagram Protocol(UDP).
  - Stream Control Transmission Protocol(SCTP).

TCP/IP

- Internetwork Layer Protocols
  - Internetworking Protocol(IP)
- No specific protocol for physical and data link layer.

Addresses in TCP/IP

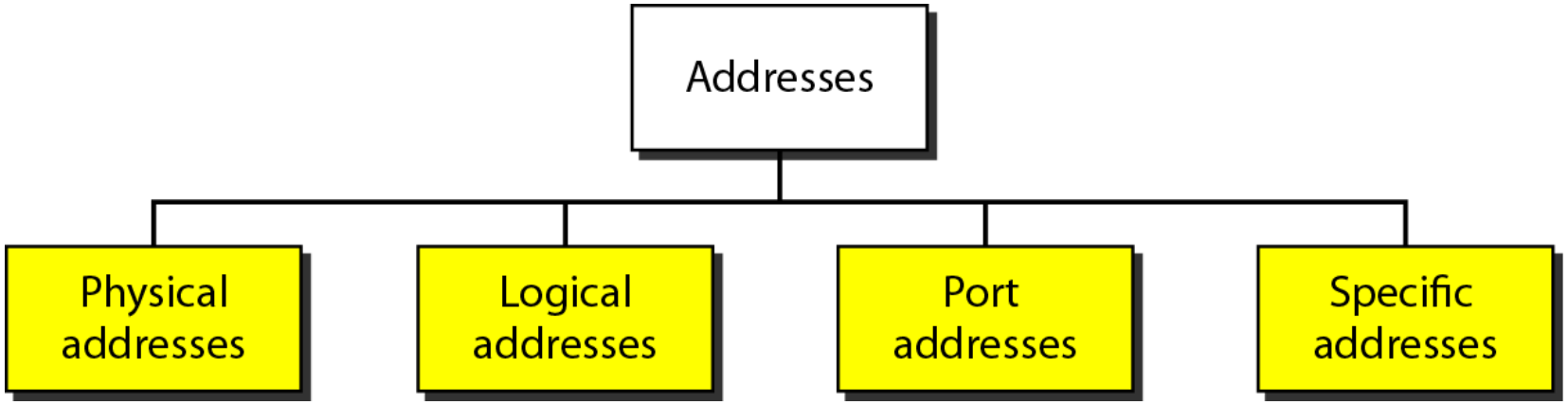

Image Courtesy: Data Communication and Networking
Behrouz A. Forouzan, McGraw-Hill

Summary

- Discussed the network architectures in computer networks and the two types of network models; P2P model and Client/Server model.

References

1 Data Communication and Networking Behrouz A. Forouzan, McGraw-Hill.

2 J. Glenn Brookshear,"Computer Science: An Overview", Addision-Wesley, Twelth Edition, 2014